

Mean value of a function

Key Points

The mean value of $f(x)$ over the interval $[a, b]$ is

$$\frac{1}{b-a} \int_a^b f(x) dx$$

It represents the average area per unit in x (over the interval)/ The average $\bar{y}$

Basic Skills

Q1. Compute the following integrals

a) $\int \sqrt[5]{x^4} dx$

b) $\int e^x - \frac{1}{x} dx$

c) $\int \sin x - (3x + 1)^5 dx$

Q2. Find the area under the curve $y = \sqrt{x}$ between $x = 0$ and $x = 9$

Q3. For the function $f(x) = x^2$

a) Find the area of the function over the interval $[1, 5]$

b) Hence find the mean area over the interval $[1, 5]$

Q4. Find the mean value of $f(x) = \sqrt{x}$ over the interval $[4, 16]$

Competency Skills

Q5. For the curve $y = 9 - x^2$ over the interval $[-3, 3]$

a) Find the mean value function over the given interval

b) Sketch the curve over the interval and indicate the mean value

Q6. Find the mean value of the following functions

a) $f(x) = e^x$ over the interval $[0, 2]$

b) $f(x) = \frac{1}{\sqrt{x}}$ over the interval $[1, 4]$

c) $f(x) = \frac{1}{x}$ over the interval $[1, 2]$

Q7. Show that the mean value of function $f(x) = (2x - 4)^4$ is the same over the intervals $[0, 2]$ and $[0, 4]$

Advanced Skills

Q8. Given that the mean value of $y = ax^2$ over $[0, b]$ and $y = bx^2$ over $[0, a]$ are the same, find the possible values of a, b

Q9. For the function $f(x) = \sin x$

a) Find the mean value over the period for the function

b) Another type of mean is the root mean square $\sqrt{\frac{1}{b-a} \int_a^b (f(x))^2 dx}$, find this root mean square over the period for the function

c) Compare your answers to part a) and b)

Q10. Given a function has a mean of 2 over $[0, 4]$ and 10 over $[4, 6]$ find the mean over $[0, 6]$

Extension

Q11. Explain the difference between the mean value of a function and the mean value of a pdf